ALL INDIA TEST SERIES

FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

CONCEPT RECAPITULATION TEST - III

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

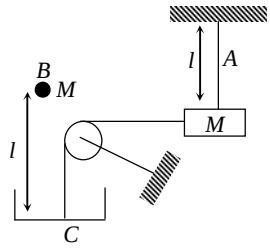
PART – I

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

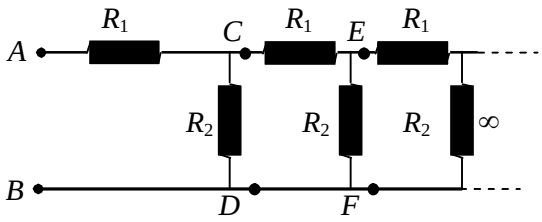
1. In the figure shown, pan *C* is massless. All strings and pulleys are ideal. Block *B* is dropped from a height *l* on pan *C*. Collision between block *B* and pan *C* is perfectly inelastic. Just after collision tension in string *A* differ from tension in string *A* before collision by a magnitude of



(A) Mg (B) $\frac{Mg}{2}$

(C) $\frac{Mg}{4}$ (D) 2Mg

2. An infinite grid is made using resistor R_1 and R_2 as shown. The potential difference between *A* and *B* is *V*, between *C* and *D* is $\frac{V}{3}$, between *E* and *F* is $\frac{V}{9}$ i.e. potential difference becomes one third after each pair. The value of $\frac{R_1}{R_2}$ is

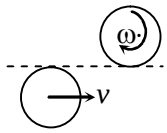


(A) 1 (B) $\frac{4}{3}$

(C) $\frac{2}{3}$ (D) 2

Space for rough work

3. A uniform disc of mass m and radius *R* is rotating with angular velocity ω on a smooth horizontal surface. Another identical disc is moving translationally with velocity *v* as shown.



When they touch each other, they stick together. The angular velocity of centre of mass of the system after contact will be

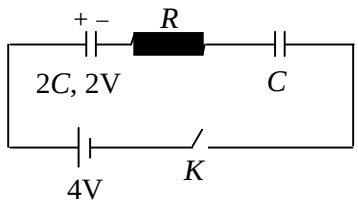
(A) zero

(B) $\frac{2v - R\omega}{6R}$

(C) $\frac{v}{R}$

(D) $\frac{\omega}{6}$

4. A capacitor of capacitance 2*C* is charged to a potential 2V and then connected to an uncharged capacitor of capacitance *C* with a resistance *R* to a battery of emf 4V as shown.
The key is closed at $t = 0$. The charge on uncharged capacitor as a function of time will be



(A) $4CV\left(1 - e^{\frac{-t}{2RC}}\right)$

(B) $\frac{4CV}{3}\left(1 - e^{\frac{-t}{2RC}}\right)$

(C) $4CV\left(1 - e^{\frac{-3t}{2RC}}\right)$

(D) $\frac{4CV}{3}\left(1 - e^{\frac{-3t}{2RC}}\right)$

5. A series circuit consisting of a capacitor and a coil with active resistance is connected to a source of harmonic voltage whose frequency can be varied, keeping the voltage amplitude constant. At frequencies ω_1 and ω_2, the current amplitudes are *n* times less than the resonance amplitude. Then resonance frequency will be

(A) $\frac{\omega_1 + \omega_2}{2}$

(B) $\omega_1 + \omega_2$

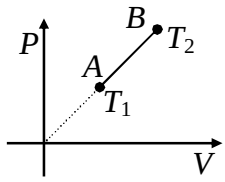
(C) $\omega_1 - \omega_2$

(D) $\sqrt{\omega_1\omega_2}$

6. If the tension in the string changes by 21%, the fundamental frequency of string changes by 15 Hz. Which of the following statement is correct?
(A) original fundamental frequency is nearly 150 Hz
(B) velocity of propagation changes by 4.5%
(C) velocity of propagation changes by 10%
(D) fundamental wavelength changes by 15%

Space for rough work

7. Two moles of an ideal monatomic gas are taken from state *A* to state *B* through a process as shown in figure. The work done by the gas in the process is

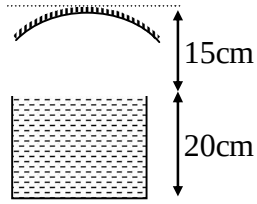


(A) $R(T_2 - T_1)$ (B) $2R(T_2 - T_1)$

(C) $\frac{1}{2}R(T_2 - T_1)$ (D) $\frac{1}{4R}(T_2 - T_1)$

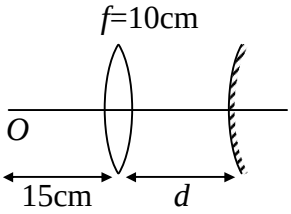
8. When an ideal gas undergoes an isothermal expansion, the pressure of the gas decreases due to
(A) decrease in change in momentum per collision
(B) decrease in frequency of collision with the container
(C) both decrease in frequency of collision and change in momentum per collision
(D) none of these

9. A vessel is filled with water $\left(\mu = \frac{4}{3}\right)$ upto a height 20 cm. A concave mirror of radius of curvature 40 cm is placed 15 cm above the water level. The position of final image of an object placed 20 cm below the water surface will be



(A) 45 cm below the water surface
(B) 60 cm above the water surface
(C) 45 cm above the water surface
(D) 60 cm below the water surface

10. An object is placed in front of convex lens of focal length (*f* = 10 cm). The distance *d* for which image of the object will concide with the object (irrespective of focal length of the mirror)



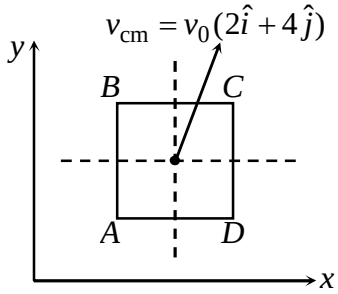
(A) 30
(B) 20
(C) 0
(D) 15

11. The probability that a radioactive nucleus will survive 10 times longer than its half life, is
(A) 2^{-20} (B) 2^{-10}
(C) $\left(\frac{1}{2}\right)^5$ (D) data insufficient

Space for rough work

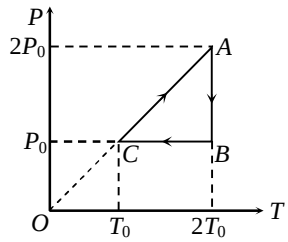
12. In a screw gauge, there are 100 divisions in circular scale and each main scale division is of 1 mm. When there is no gap between the jaws, 97th divisions coincides with the main scale zero and zero of main scale is not visible. While measuring the diameter of a ball, the circular scale is between 3 mm mark and 4 mm mark such that the 76th division of circular scale coincides with the reference line. Select the correct alternative
(A) the least count of the micrometer is 0.01 cm
(B) the zero error is – 0.04 mm
(C) the diameter of the ball is 3.79 cm
(D) the main scale reading is 4 mm

13. A uniform solid square plate *ABCD* of mass m and side *a* is moving in *x – y* horizontal smooth plane. The velocity of centre of mass is $v_0(2\hat{i}+4\hat{j})$ m/s. The end *A* of square plate is suddenly fixed by a pin, find the new velocity of centre of mass of square



(A) $v_0(2\hat{i}+4\hat{j})$
(B) $\frac{3v_0}{4}(-\hat{i})+\frac{3v_0}{4}\hat{j}$
(C) $\frac{3v_0}{2}\hat{i}-\frac{3v_0}{2}\hat{j}$
(D) $3v_0\hat{i}-3v_0\hat{j}$

14. Three moles of an ideal gas goes to a cyclic process shown in figure. The work done by the gas during the process is



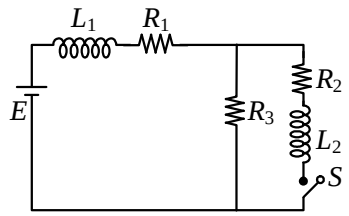
(A) $2.36RT_0$
(B) $0.58RT_0$
(C) $1.16RT_0$
(D) $-0.58RT_0$

15. A uniform ring of mass '*m*' and radius '*R*' is projected horizontally with velocity v_0 on a rough horizontal floor, so that it starts off with a purely sliding motion and it acquires a purely rolling motion after moving a distance *d*. If the coefficient of friction between the ground and ring is μ, then work done by the friction in the process is

(A) $-\mu mgd$
(B) $-\frac{1}{4}mv_0^2$
(C) μmgd
(D) $-\frac{1}{8}mv_0^2$

Space for rough work

16. In the circuit shown in figure, the switch '*S*' is closed at $t = 0$. After closing the switch, in steady state, the value of current in the resistor R_3



(A) increases
(B) decreases
(C) remains constant
(D) becomes zero

17. A particle is moving in $x - y$ plane with a constant speed v_0 such that its y displacement is given by $y = \alpha e^{\frac{-2v_x}{\sqrt{3}v_0}}$. Where v_x is component of velocity along x axis. y displacement of the particle when slope of tangent on its path is $-\frac{1}{\sqrt{3}}$, is

(A) αe^{-1} (B) αe^{-2}
(C) zero (D) $\alpha^2 e$

18. A uniform string of large length L of uniform cross section, mass per unit length λ and free at both ends is revolving round the earth at equator in vertical position in such a way that its position relative to the earth is always the same and its lower end is slightly above the earth's surface. If the string is pulled down by a small distance x, then work done by gravity is

(A) $GM\lambda x\left(\frac{R}{R+L}\right)$ (B) $\frac{GM\lambda xL}{R(R+L)}$
(C) $\frac{GM\lambda x^2}{R(R+L)}$ (D) $\frac{GM\lambda L^2}{R(R+L)}$

19. A particle on a spring executes SHM. If the mass of the particle and the amplitude are both doubled then the magnitude of the maximum acceleration of the particle will change by a factor of

(A) 4 (B) $\sqrt{8}$
(C) 2 (D) 1

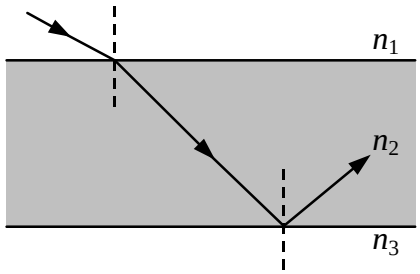
20. Two waves travel down the same string. The waves have the same velocity, frequency (f_0), and wavelength but different phase constants ($\phi_1 > \phi_2$) and amplitudes ($A_1 > A_2$)
According to the principle of superposition, the resultant wave has an amplitude A such that

(A) $A = A_1 + A_2$ (B) $A = A_1 - A_2$
(C) $A_1 \le A \le A_2$ (D) $A_1 - A_2 \le A \le A_1 + A_2$

Space for rough work

21. You are provided with three similar, but slightly different, tuning forks. When *A* and *B* are both struck, a beat frequency of f_{AB} is heard. When *B* and *C* are both struck, a beat frequency of f_{BC} is heard. It was noticed that $f_{AB} < f_{BC}$
A and C are simultaneously struck. What will be the observed beat frequency?
(A) $|f_{AB} + f_{BC}|$
(B) $|f_{AB} - f_{BC}|$
(C) either $|f_{AB} + f_{BC}|$ or $|f_{AB} - f_{BC}|$ will be heard
(D)both $|f_{AB} + f_{BC}|$ and $|f_{AB} - f_{BC}|$ will simultaneously be heard

22. Light traveling through three transparent substances follows the path shown in figure. Arrange the indices of refraction in order from smallest to largest. Note that total internal reflection does occur on the bottom surface of medium 2



(A) $n_1 < n_2 < n_3$
(B) $n_2 < n_1 < n_3$
(C) $n_1 < n_3 < n_2$
(D) $n_3 < n_1 < n_2$

23. Light shines through a spherical air bubble underwater. What type of optical device does the bubble act like?
(A) Converging
(B) Diverging
(C) Planar
(D) none of these

24. For fun, a student constructs a three-slit interference experiment. If the light from each of the three slits arrives in phase at the central maximum, then how will the intensity *I* compare to the intensity I_0 from a single slit?
(A) $I = I_0$
(B) $I = 3I_0$
(C) $I = 6I_0$
(D) $I = 9I_0$

25. An ideal gas has an initial pressure of 3 units and an initial volume of 4 units. The table gives the final pressure and volume of the gas (in same unit) in five processes. Which processes start and end on the same isotherm?

	a	b	c	d	e
P	12	6	5	4	2
V	1	2	7	3	12

(A) a, b, d
(B) a, b, d, e
(C) only c
(D) d, e

Space for rough work

26. If two moles of diatomic gas and one mole of mono atomic gas are mixed then the ratio of specific heats is

(A) $\frac{7}{3}$ (B) $\frac{5}{4}$

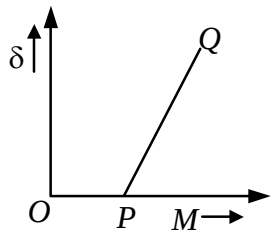
(C) $\frac{19}{13}$ (D) $\frac{15}{19}$

27. The K_α wavelength of an element with atomic number $Z = 17$ is given by λ. What is the atomic number of an element with a K_α wavelength of 4λ?

(A) 4 (B) 5

(C) 7 (D) 9

28. For a small angled prism, angle of prism A, the angle of minimum deviation (δ) varies with the refractive index of the prism as shown in the graph



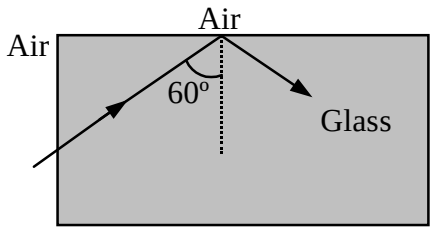
(A)point P corresponds to $\mu > 1$

(B)slope of the line $PQ = A/2$

(C)slope = A

(D)none of the above statements is true

29. A light ray from air is incident (as shown in figure) at one end of a glass fiber (refractive index $\mu = 1.5$) making an incidence angle of 60º on the lateral surface, so that it undergoes a total internal reflection. How much time would it take to traverse the straight fiber of length 1 km



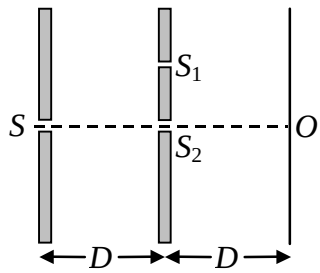
(A) 3.33 μ sec

(B) 6.67 μ sec

(C) 5.77 μ sec

(D) none of these

30. Two ideal slits S_1 and S_2 are at a distance d apart, and illuminated by light of wavelength λ passing through an ideal source slit S placed on the line through S_2 as shown. The distance between the planes of slits and the source slit is D. A screen is held at a distance D from the plane of the slits. The minimum value of d for which there is darkness at O is



(A) $\sqrt{\frac{3\lambda D}{2}}$ (B) $\sqrt{\lambda D}$

(C) $\sqrt{\frac{\lambda D}{2}}$ (D) $\sqrt{3\lambda D}$

Space for rough work

Chemistry

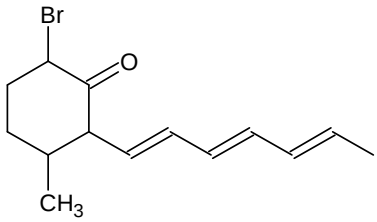
PART – II

SECTION – A

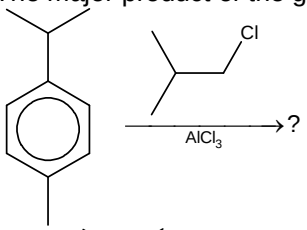
Straight Objective Type

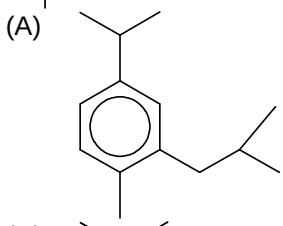
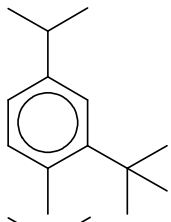
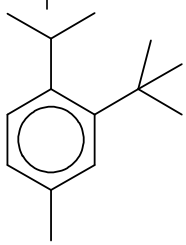
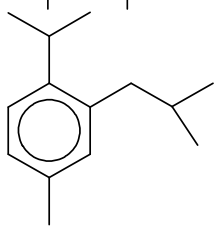
This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. The basicity of Aliphatic Amines in gas phase is
   (A) 1° amine > 2° amine > 3° amine
   (B) 3° amine > 2° amine > 1° amine
   (C) 2° amine > 1° amine > 3° amine
   (D) 2° amine > 3° amine > 1° amine

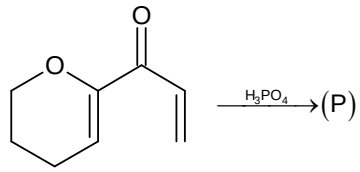
2. 

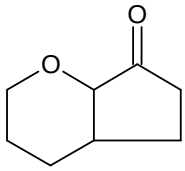
How many stereo isomers are possible for above compound?
   (A) 14
   (B) 16
   (C) 12
   (D) 18

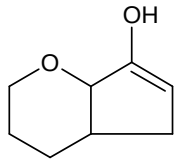
3. The major product of the given reaction is


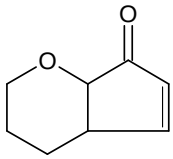
   (A) 
   (B) 
   (C) 
   (D) 

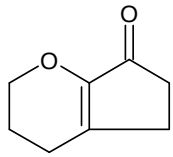
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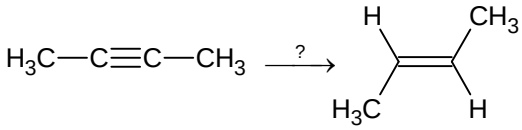
4. 

(A) 

(B) 

(C) 

(D) 

5. 

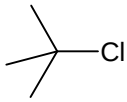
(A) $NH_2 - NH_2 \,|\, OH^-$

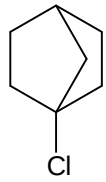
(B) $H_2 \,|\, Ni$

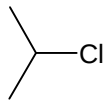
(C) $H_2 \,|\, Pd - BaSO_4$

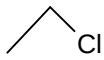
(D) $H_2 \,|\, Ll.\ liq\ NH_2$

6. Which of the following halide readily undergo S_N1 reaction?

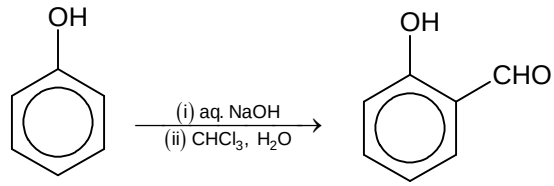
(A) 

(B) 

(C) 

(D) 

Space for rough work

7. 

The intermediate involved is:

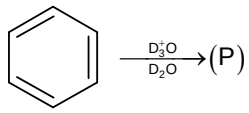
(A) $:CH_2$ (B) $:CCl_2$

(C) $\overset{\oplus}{C}H_3$ (D) $\overset{\oplus}{C}Cl_2$

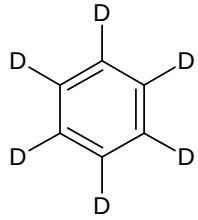
8. When accetaldehyde is heated with Fehling's solution. It gives a precipitate of

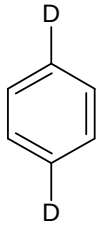
(A) Cu (B) CuO

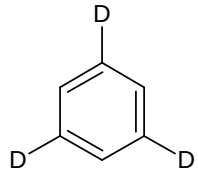
(C) Cu_2O (D) $Cu + Cu_2O + CuO$

9. 

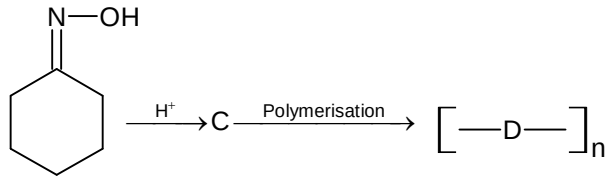
The product (P) formed finally is?

(A) 

(B) 

(C) 

(D) None of these

10. 

What is D?

(A) Nylon-66 (B) Nylon – 6

(C) Bekalite (D) Buna – 9

Space for rough work

11. The electronic configuration of four elements are
(1) $[Xe]6s^1$
(2) $[Xe]4f^{14}, 5d^1, 6s^2$
(3) $[Ar]4s^2 4p^5$
(4) $[Ar]3d^7, 4s^2$
Which one of the following statements about these elements is not correct?
(A) 1 is strong reducing agent.
(B) 2 is a d-block element.
(C) 3 has high electron affinity.
(D) The compound formed between 1 and 3 is ionic.

12. The IP_1, IP_2, IP_3, IP_4 and IP_5 of an element are 7.1, 14.3, 34.5, 46.8, 162.2 eV respectively. The element is likely to be
(A) Na (B) Si
(C) F (D) Ca

13. H_2S is bubbled into 0.2 M NaCN solution which is 0.02 M in each $[Cd(CN)_4]^{2-}$ and $[Ag(CN)_2]^-$. The H_2S produces 1×10^{-9} M sulphide ion in the solution.
Given: $K_{sp}Ag_2S = 1\times10^{-50}\ M^3; K_{sp}CdS = 7.1\times10^{-28}\ M^2$
$K_{instability}\left[Ag(CN)_2\right]^{-1} = 1\times10^{-20}\ M^2, K_{instability}\left[Cd(CN)_4\right]^{2-} = 7.8\times10^{-18}\ M$
Identify the correct statement:
(A) Ag_2S precipitates first from the solution.
(B) CdS precipitates first from the solution.
(C) None of them precipitates under the given conditions.
(D) Ag_2S precipitates at a sulphide concentration $1 \times 10^{+15}$ M.

14. Extraction of Ag from commercial lead is done by
(A) Parke's Process (B) Clarke's Process
(C) Kroll's Process (D) Electrolytic Process

15. Most stable polymeric hydride among the following is:
(A) CaH_2 (B) MgH_2
(C) BaH_2 (D) SrH_2

Space for rough work

16. $NaOH + O_3 \longrightarrow X + \text{products}$

X is

(A) Na_2O (B) NaO_3

(C) NaO_2 (D) Na_2O_2

17. Arrange the bond angle order for Cl_2O, ClO_2, F_2O, Br_2O:

(A) $OF_2 > OCl_2 > OBr_2 > ClO_2$ (B) $OF_2 < OCl_2 < OBr_2 < ClO_2$

(C) $ClO_2 < OF_2 < OCl_2 < OBr_2$ (D) $OF_2 < OCl_2 < ClO_2 < OBr_2$

18. A d-Block element forms octahedral complex but its magnetic moment remains same either in strong field or in weak field ligand. Which of the following is/are correct?

(A) Element always forms colourless compound.

(B) Number of electrons in t_{2g} orbitals are higher than in e_g orbitals.

(C) It can have either d^3 or d^8 configuration.

(D) It can have either d^7 or d^8 configuration.

19. Among the following complexes which will be attracted by magnetic field?

$\underset{(I)}{\left[Ni(CN)_4\right]^{2-}}$, $\underset{(II)}{\left[NiCl_4\right]^{2-}}$, $\underset{(III)}{\left[Ni(CO)_4\right]}$, $\underset{(IV)}{\left[Ni(H_2O)_6\right]^{2+}}$

(A) (I) only (B) (I) and (IV)

(C) (II), (III) and (IV) (D) (II) and (IV)

20. H_2S gas on passing through $CdSO_4$ solution will give

(A) Black ppt (B) Yellow ppt

(C) Organge ppt (D) No ppt

21. The vapour density of a mixture containing NO_2 and N_2O_4 is 27.6. The mole fraction of N_2O_4 in the mixture is

(A) 0.1 (B) 0.2

(C) 0.5 (D) 0.8

22. Calculate the maximum number of electrons which may have magnetic quantum number m = + 1 and spin quantum number s = +1/2 in chromium.

(A) 0 (B) 1

(C) 4 (D) 3

23. 36.5% HCl has density equal to 1.2 g ml^{-1}. The molarity (M) and molality (m), respectively are

(A) 15.7, 15.7 (B) 12, 12

(C) 15.7, 12 (D) 12, 15.7

Space for rough work

24. $3KClO_3 + 3H_2SO_4 \longrightarrow 3KHSO_4 + HClO_4 + 2ClO_2 + H_2O$

Equivalent weight of $KClO_3$ is

(A) $\frac{M}{4}$ (B) $\frac{M}{2}$

(C) $\left(M + \frac{M}{2}\right)$ (D) $\left(\frac{M}{4} + \frac{M}{2}\right)$

25. An ideal gas obeying kinetic theory of gases can be liquefied, if

(A) It's temperature is more than critical temperature T_c.

(B) Its pressure is more than critical pressure P_c.

(C) It's pressure is more than P_c at a temperature less than T_C.

(D) It cannot be liquefied at any value of P and T

26. A sample of Argon gas at 1 atm pressure and 27°C expands reversibly and adiabatically from 1.25 dm^3 to 2.5 dm^3. Calculate the enthalpy. Change in this process. C_{vm} for Argon is 12.48 $Jk^{-1}mol^{-1}$

(A) –114.52 J (B) –110.32 J

(C) –100.42 J (D) –126.42 J

27. In reduction

$\underset{(g)}{CH_3COCH_3} \rightleftharpoons \underset{(g)}{CH_3CH_3} + CO(g)$, if the initial pressure of $CH_3COCH_3(g)$ is 150 mm and at equilibrium the mole fraction of CO(g) is $\frac{1}{3}$, then the value K_p is

(A) 50 mm (B) 100 mm

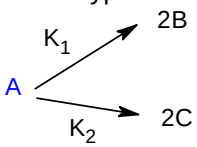
(C) 33.3 mm (D) 75 mm

28. Calculate pH of saturated solution of $Mg(OH)_2$, K_{sp} for $Mg(OH)_2$ is 8.9×10^{-12}

(A) 9.4 (B) 11.2

(C) 10.4 (D) 9.8

29. For a hypothetical elementary reaction:



Where $\frac{K_1}{K_2} = \frac{1}{2}$

Initially only 2 moles of A are present. The total no. of moles of A, B and C at the end of 75% reaction are

(A) 2 (B) 3

(C) 4 (D) 3.5

30. A crystal is made of particles X and Y. X forms f_{cc} packing and Y occupies all the octahedral voids. If all the particles along one body diagonal are removed then the formula of the crystal would be:

(A) X_4Y_3 (B) X_5Y_4

(C) X_4Y_5 (D) None of these

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. If α, β are the roots of the equation $x^2 + px + q = 0$ and also of the equation $x^{2n} + p^n x^n + q^n = 0$ and if $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of the equation $x^n + 1 + (x + 1)^n = 0$ then n must be an
   (A) even integer (B) odd integer
   (C) irrational number (D) none of these

2. A pole 50m high stands on a building 250 m high to an observer at a height of 300 m, the building and the pole subtend equal angles. The distance of the observer from the top of the pole is
   (A) 25 m (B) 50 m
   (C) $25\sqrt{3}$m (D) $25\sqrt{6}$m

3. Let $P = \prod_{n=1}^{\infty}\left(10^{\frac{1}{2^{n-1}}}\right)$ then $\log_{0.01}(P)$ is
   (A) 0 (B) –1
   (C) 1 (D) none of these

4. The greatest value of the function $y = \frac{x}{ax^2 + b}$ (a, b > 0)
   (A) $\frac{1}{\sqrt{ab}}$ (B) $\frac{1}{2\sqrt{ab}}$
   (C) $\sqrt{ab}$ (D) $2\sqrt{ab}$

Space for rough work

5. If a variable takes the values 0, 1, 2, ….., n with frequencies 1, ${}^nC_1$, ${}^nC_2$, …, ${}^nC_n$, then the A.M. is

(A) n (B) $\frac{2^n}{n}$

(C) n + 1 (D) $\frac{n}{2}$

6. The greatest coefficient in the expansion of $(a + b + c + d)^{15}$ is

(A) $\frac{15!}{(3!)^2(4!)^3}$ (B) $\frac{15!}{(3!)(4!)^3}$

(C) $\frac{15!}{(3!)^3(4!)^2}$ (D) none of these

7. Let S be the set of all triangles and R^+ be the set of positive real numbers, then the function, $f: S \to R^+$, $f(\Delta)$ = area of Δ, where $\Delta \in S$ is

(A) injective but not surjective (B) surjective but not injective

(C) injective as well as surjective (D) neither injective nor surjective

8. Tangents are drawn to $x^2 + y^2 = 16$ from P(0, h). These tangents meet x-axis at A and B. If the area of PAB is minimum, then value of h is

(A) $12\sqrt{2}$ (B) $8\sqrt{2}$

(C) $4\sqrt{2}$ (D) $\sqrt{2}$

9. If $\alpha + \beta = 3\pi$ then the chord joining the points α and β for the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through

(A) focus (B) centre

(C) one of the end points of the transverse axis (D) one of the end points of the conjugates axes

10. For a given non-zero value of m each of the lines $\frac{x}{a} - \frac{y}{b} = m$ and $\frac{x}{a} + \frac{y}{b} = m$ meets the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at a point, sum of the ordinates of these points, is

(A) $\frac{a(1+m^2)}{m}$ (B) $\frac{b(1-m^2)}{m}$

(C) 0 (D) $\frac{a+b}{2m}$

Space for rough work

11. Two points P and Q are given in the rectangular cartesian co-ordinates on the curve $y = 2^{x+2}$, such that $\overrightarrow{OP} \cdot \hat{i} = -1$ and $\overrightarrow{OQ} \cdot \hat{i} = 2$ where $\hat{i}$ is a unit vector along the x-axis. Then the magnitude of $\overrightarrow{OQ} - 4\overrightarrow{OP}$ is
(A) 8 (B) 10
(C) 12 (D) 16

12. The image of A(p, q) in the line y = –x is B and the image of B in the line y = x is C then the mid point of AC is
(A) (p + q, p + q) (B) $\left(\frac{p+q}{2}, \frac{p+q}{2}\right)$
(C) (p – q, q – p) (D) (0, 0)

13. Let PQR be a triangle let P be the point (1, 2), y = x is the perpendicular bisector of PQ and x – 2y + 1 = 0 is the angel bisector of $\angle R$. If equation of QR is given by $\lambda x + \mu y - 5 = 0$, then the value of $\lambda + \mu$ is
(A) 1 (B) 2
(C) 3 (D) 4

14. If the tangents are drawn from any point on the line x + y = 3 to the circle $x^2 + y^2 = 9$, then the chord of contact passes through the point
(A) (3, 5) (B) (3, 3)
(C) (5, 3) (D) none of these

15. A is a point (a, b) in the first quadrant. If the two circles which passes through A and touches the coordinate axes cut at right angles then
(A) $a^2 - 6ab + b^2 = 0$ (B) $a^2 + 2ab - b^2 = 0$
(C) $a^2 - 4ab + b^2 = 0$ (D) $a^2 - 8ab + b^2 = 0$

16. If P be a point on the parabola $y^2 = 3(2x - 3)$ and M is the foot of perpendicular drawn from P on the directrix of the parabola, then length of each sides of an equilateral triangle SMP (where S is the focus of the parabola), is
(A) 2 (B) 4
(C) 6 (D) 8

17. If the chord of contact of tangents from a point P to the parabola $y^2 = 4ax$ touches the parabola $x^2 = 4by$, then locus of P is
(A) circle (B) parabola
(C) ellipse (D) hyperbola

Space for rough work

18. A tangent to the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ at any points meet the line x = 0 at a point Q. Let R be the image of Q in the line y = x, then circle whose extremities of a diameter are Q and R passes through a fixed point, the fixed point is
(A) (3, 0) (B) (5, 0)
(C) (0, 0) (D) (4, 0)

19. Number of distinct normal lines that can be drawn to ellipse $\frac{x^2}{169}+\frac{y^2}{25}=1$ from the point P(0, 6) is
(A) 1 (B) 2
(C) 3 (D) 4

20. The conditional $(p \wedge q) \Rightarrow p$ is
(A) a tautology (B) a contradiction
(C) neither a tautology nor a contradiction (D) none of these

21. If f(x) = (x – 1)(x – 2)(x – 3)(x – 4), then out of the three roots of f′(x) = 0
(A) three are positive (B) three are negative
(C) two are complex (D) none of these

22. Let f(x) g(y) = g′(y) – f′(x) g(y) $\forall$ x, y $\in$ R and g′(0) = 1, g(0) = 1, f′(0) = –5, then
(A) f(0) = 8 (B) f(1) = e
(C) g(1) = e (D) $g(-1) = e^{-1}$

23. At an election, a voter may vote for any number of candidates not greater than the number to be elected, there are 10 candidates and 4 are to be elected. The number of ways in which a voter can vote for atleast one candidate is
(A) 385 (B) 1110
(C) 5040 (D) 4050

24. The number of 5 digit numbers that can be made using the digits 1 and 2 and in which at least one digit in different, is
(A) 30 (B) 31
(C) 32 (D) 35

Space for rough work

25. If two numbers a and b are chosen from the set of integers 1, 2, 3, 4, ….., 39 then, the probability that the equation 7a – 9b = 0 is satisfied, is

(A) $\frac{1}{247}$ (B) $\frac{2}{247}$

(C) $\frac{4}{741}$ (D) $\frac{5}{741}$

26. Two numbers are selected randomly from set S = {1, 2, 3, 4, 5, 6} without replacement one by one the probability that minimum of the two number is less than 4, is

(A) $\frac{1}{15}$ (B) $\frac{14}{15}$

(C) $\frac{1}{5}$ (D) $\frac{4}{5}$

27. If $\int \frac{dx}{\sqrt{\sin^3 x \cos x}} = g(x) + c$, then g(x) is equal to

(A) $-\frac{2}{\sqrt{\cos x}}$ (B) $-\frac{2}{\sqrt{\tan x}}$

(C) $\frac{2}{\sqrt{\cot x}}$ (D) $\frac{2}{\sqrt{\tan x}}$

28. The number of real solution of $6^x + 1 = 8^x - 27^{x-1}$ is/are

(A) exactly one (B) exactly two

(C) more than two (D) none of these

29. The value of $\theta \in (0, 2\pi)$ for which the equation is $2\sin^2\theta - 5\sin\theta + 2 > 0$, is

(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$ (B) $\left(\frac{\pi}{8}, \frac{5\pi}{6}\right)$

(C) $\left(0, \frac{\pi}{8}\right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$ (D) $\left(\frac{41\pi}{48}, \pi\right)$

30. If f: R $\to$ R and g: R $\to$ R are two function such that $f(x) + f''(x) = -xg(x)f'(x)$ and $g(x) > 0 \ \forall \ x \in R$ then $f^2(x) + (f'(x))^2$ has

(A) maxima at x = 0 (B) min at x = 0

(C) point of inflexion at x = 0 (D) none of these

Space for rough work